

M33. Interferences in the Quincke Tube

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Abstract

This report investigates the phenomenon of acoustic interference using a Quincke tube. By measuring the transmitted sound intensity as a function of both frequency and path length difference, constructive and destructive interference patterns were identified. These observations were used to determine the speed of sound and the characteristic length of the apparatus. The experimental results for the speed of sound were compared with theoretical values calculated from the ambient temperature.

1 Introduction and Objectives

1.1 Objectives

- Verify the occurrence of constructive and destructive interference in sound waves.
- Determine the speed of sound through the analysis of interference patterns.
- Calculate the characteristic length of the Quincke tube.

1.2 Theoretical Background

Interference occurs when two or more waves coincide in space and time. In a Quincke tube, a pressure wave from a source is split into two paths of lengths a and b . These waves interfere at a receiver. The path difference is given by:

$$b - a = 2\Delta d \quad (1)$$

where Δd is the displacement of the movable arm. The condition for constructive interference (maxima) and destructive interference (minima) is:

$$\Delta d = n \frac{c}{4f} \quad (2)$$

where n is a relative ordinal, c is the speed of sound, and f is the frequency. When the paths are equal ($a = b$), resonance frequencies can occur according to:

$$f_n = n \frac{c}{2a} \quad (3)$$

2 Materials and Methods

2.1 Experimental Setup

The setup includes a Quincke tube with a movable arm, a speaker as a sound source connected to a function generator, and a microphone connected to a voltmeter or oscilloscope.

2.2 Data Acquisition and Analysis

Two main experiments were performed:

1. **Experiment A/B:** Frequencies corresponding to maxima and minima were identified for a fixed tube configuration ($a \approx b$).

2. **Experiment C:** The sound amplitude (voltage) was recorded as a function of the arm displacement Δd for a fixed frequency $f = 1576$ Hz.

The speed of sound c was determined by fitting the positions of maxima and minima in Experiment C to the linear relation $\Delta d(n)$.

3 Results and Data Analysis

3.1 Experimental Data

The direct measurements obtained during the laboratory session are presented below. Table 1 displays the voltage recorded at various frequencies to identify the resonance modes (Experiment A). Table 2 displays the voltage measured as a function of the movable arm's displacement Δd at a constant frequency of $f = 1576$ Hz (Experiment C).

Note: The uncertainty for the voltage is calculated as $u_V = 0.05 \cdot V + 0.001$ V. The uncertainty for the frequency is $u_f = 0.05 \cdot f$. The uncertainty for the displacement is $u_{\Delta d} = 0.001$ m.

Table 1: Voltage vs. Frequency (Experiment A)

Frequency f (Hz)	Voltage V (V)	u_f (Hz)	u_V (V)
1280	1.3	64	0.07
1380	6.9	69	0.35
1460	0.9	73	0.05
1576	10.7	79	0.54
1660	4.1	83	0.21
1710	2.7	86	0.14
1810	1.7	91	0.09
1890	1.4	95	0.07
2000	1.0	100	0.05
2070	4.4	104	0.22
2120	1.3	106	0.07
2190	2.1	110	0.11

Table 2: Voltage vs. Displacement at $f = 1576$ Hz (Experiment C) - Extracted Extrema

Extrema Type	Order (n)	Displacement Δd (m)	Voltage V (V)
Minimum	1	0.045	0.5
Maximum	1	0.078	2.6
Minimum	2	0.103	0.4
Maximum	2	0.110	14.3
Minimum	3	0.185	0.4
Maximum	3	0.230	14.1
Minimum	4	0.245	0.8
Maximum	4	0.310	10.6

Oscilloscope Captures

Figure 1 shows the raw signals observed on the oscilloscope for a state of constructive interference (maximum amplitude) and destructive interference (minimum amplitude).

Figure 1: Oscilloscope captures showing constructive (left) and destructive (right) interference.

3.2 Voltage vs Frequency

The experimental results for the transmitted sound intensity as a function of frequency are displayed in Figure 2.

Figure 2: Sound amplitude (Voltage) as a function of source frequency.

3.3 Voltage vs Distance

For a fixed frequency $f = 1576$ Hz, the voltage was measured as the arm displacement Δd was varied. The resulting interference pattern is shown in Figure 3.

Figure 3: Sound amplitude (Voltage) as a function of movable arm displacement Δd .

3.4 Determination of Sound Speed

The detected peaks (maxima) and troughs (minima) from Experiment C were used to perform a linear fit of Δd vs n .

Figure 4: Linear fits for sound speed determination using maxima (left) and minima (right).

The resulting values for the speed of sound are:

- **From Maxima:** $c = 380.54 \pm 0.85$ m/s
- **From Minima:** $c = 374.02 \pm 1.2$ m/s

The theoretical speed of sound at $\theta = 27.5^\circ\text{C}$ (equivalent to $T = 300.65$ K) is calculated as:

$$c_{theory} = 331.3 \sqrt{1 + \frac{\theta}{273.15}} = 331.3 \sqrt{\frac{300.65}{273.15}} = 347.58 \text{ m/s} \quad (4)$$

The relative differences between the experimental and theoretical values are:

- **Maxima:** 9.48 %
- **Minima:** 7.61 %

3.5 Determination of the Characteristic Length (a)

According to the theoretical model for a closed tube (Equation 3), the resonance frequencies f_n are linearly dependent on the mode order n :

$$f_n = n \frac{c}{2a} \quad (5)$$

By plotting the resonance frequencies (maxima from Experiment A) as a function of their corresponding relative order n , a linear fit $f_n = m \cdot n$ can be performed. The slope m of this fit is directly related to the characteristic length of the tube a :

$$m = \frac{c}{2a} \implies a = \frac{c}{2m} \quad (6)$$

Using the experimental speed of sound obtained from the maxima fit ($c = 380.54$ m/s) and the slope m from the frequency fit (e.g., $m = [\text{INSERT SLOPE}]$ Hz), the experimental value for the characteristic length of the Quincke tube is calculated as:

$$a = \frac{380.54}{2 \cdot m} = [\text{INSERT RESULT}] \text{ m} \quad (7)$$

4 Discussion and Conclusions

In this experiment, the phenomena of constructive and destructive interference of acoustic waves were successfully observed using a Quincke tube. The periodic variation of the transmitted sound intensity as a function of the path length difference (Δd) clearly demonstrated the wave nature of sound and the principle of superposition.

The speed of sound was determined experimentally using two sets of data: the positions of the interference maxima and the interference minima. The linear fits yielded $c = 380.54 \pm 0.85$ m/s (from maxima) and $c = 374.02 \pm 1.2$ m/s

(from minima). Both values are higher than the theoretical speed of sound at the laboratory temperature of 27.5°C , which was calculated to be 347.58 m/s . The relative errors were 9.48% and 7.61% , respectively.

While the order of magnitude is correct and confirms the theoretical framework, the deviation suggests the presence of systematic errors in the experimental setup. These discrepancies could stem from several sources:

- **Temperature Fluctuations:** The theoretical calculation assumes a uniform and constant temperature of 27.5°C inside the entire tube. Localized heating from the sound source or handling of the apparatus could have altered the density of the air, thus increasing the actual speed of sound.
- **Scale Calibration and Reading Errors:** The movable arm's scale may have an inherent offset, and the manual adjustment of the tube to find the exact acoustic extrema is subject to human perception and mechanical play, which increases the uncertainty in the Δd measurements.
- **Non-Ideal Wave Propagation:** The equations assume perfect plane waves. However, the presence of the T-junctions (splitters), the friction against the inner walls of the tube, and natural attenuation can introduce phase shifts and boundary effects that slightly alter the effective path length and the interference pattern.

Finally, the characteristic length of the tube a was determined by analyzing the resonance frequencies. Despite the deviations in the exact value of c , the linear trends observed in both V vs. f and V vs. Δd graphs provide robust validation of the acoustic interference model.

A Uncertainty Analysis

The uncertainty in voltage u_V was calculated as 5% of the reading plus an instrumental offset of 0.001 V . The uncertainty in frequency u_f was estimated as 5% of the set value. For the distance Δd , a fixed uncertainty of 0.1 cm was assumed based on the scale of the movable arm. The uncertainty in the fitted speed of sound was derived from the variance of the weighted linear fit through the origin.